Crowdsourced Civic Issue Reporting and Resolution System

A Project Report Submitted by

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in partial fulfillment for the award of the degree of

Bachelor of Technology

in

Information Technology

Faculty of Engineering And Technology

Marwadi University, Rajkot

2025-26

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DECLARATION

We hereby declare that the Project report submitted along with the Project entitled **Crowdsourced Civic Issue Reporting and Resolution System** for Collages submitted in partial fulfillment for the degree of Bachelor of Technology in Information Technology to Faculty of Engineering and Technology, Marwadi University, Rajkot, is a Bonafide record of original project work carried out by us at **Marwadi University** under the supervision of **Dr. Sagar Khirsariya** and that no part of this report has been directly copied from any student's reports or taken from any other source, without providing due reference.

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Institute's Vision and Mission

Institute's Vision

To foster an environment that empowers people, organisations and societies through education, ideas, research and training.

Institute's Mission

- To provide quality education and thereby bring social transformation.
- To create leaders through innovation and entrepreneurship.
- To cultivate the culture of research advancements.
- To imbibe universal consciousness.
- To stimulate growth through industrial and international partnerships.

Department’s Vision and Mission

Department’s Vision

To be recognized as a team delivering educational excellence that advances teaching, learning, and research, in alignment with Marwadi Education Foundation's mission and goals.

Department’s Mission

- To impart knowledge and skills related to the undergraduate program offered by the department.
- To impart technical and professional skills to make graduates competitive and capable.
- To constantly encourage and motivate graduates for innovation, entrepreneurship & industry readiness.
- To inspire graduates for higher education and research and to place graduates in leading industries and companies.

PEO, PO, and PSO

Program Educational Objectives (PEO):

Our graduated students are expected to fulfill the following Program Educational Objectives (PEOs):

1. **Core Competency**: Successfully apply fundamental mathematical, scientific, and engineering principles in formulating and solving engineering and real life problems for betterment of society.
2. **Breadth**: Will apply current industry accepted practices, new and emerging technologies to analyse, design, implement and maintain state of art solutions.
3. **Professionalism**: Work effectively and ethically in ever changing global professional environment and multi-disciplinary environment.
4. **Learning Environment**: Demonstrate excellent communication and soft skills to fulfil their commitment towards social responsibilities and foster life-long learning.
5. **Preparation**: Promote research and patenting to enhance technical and entrepreneurship skills within them.

Program Outcomes (POs)

Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1. Graduates will be able to identify, analyze and solve the real time problems of the industries in the area of software development, embedded system, VLSI design, IoT and communication technologies.

PSO2. Graduates will be able to contribute as an analyst and developer in the areas related to cloud computing, DevOps, security, machine learning, artificial intelligence and big

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CIRRS	Civic Issue Reporting and Resolution System
CMS	Content Management System
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DB	Database
GIS	Geographic Information System
GPS	Global Positioning System
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
ID	Identification
IT	Information Technology
JS	JavaScript
LAN	Local Area Network
ML	Machine Learning
OTP	One Time Password
PHP	Hypertext Preprocessor
SMS	Short Message Service
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
WAN	Wide Area Network

Abstract

The Crowdsourced Civic Issue Reporting and Resolution System is a user-friendly digital platform developed to bridge the gap between citizens and local authorities. It provides an easy and effective way for people to report everyday civic problems directly from their devices. Through this system, users can submit complaints related to issues such as potholes, garbage accumulation, water leakage, damaged roads, broken streetlights, and many other public concerns. Each report can include important details like location, description, and supporting images, making it easier for authorities to understand and address the problem.

The platform is designed to ensure transparency and accountability by allowing citizens to track the progress of their complaints in real time. Users can view the current status of their reported issues, whether they are pending, in progress, or resolved. This feature builds trust between the public and civic bodies by keeping communication open and visible. On the administrative side, the system provides a dedicated dashboard where officials can review submitted reports, assign tasks to staff members, update issue status, and manage the entire resolution process efficiently. By encouraging active participation from the community, the system helps create a more responsible and responsive civic environment. It reduces the need for manual complaint processes and speeds up problem resolution through organized digital management. The system is also designed to be scalable, allowing it to handle large numbers of users and be expanded to support multiple cities or regions. It can further be integrated with municipal databases and services for smoother operations.

In the future, the platform can be enhanced with modern technologies such as AI-based chatbots for user assistance, automated categorization of issues, smart suggestions based on previous reports, and advanced data analytics for better decision-making. These improvements will help authorities identify problem trends, plan civic development more effectively, and provide faster services to citizens. Overall, the system aims to create a smarter, more connected, and more efficient way of solving civic issues through technology and community collaboration.

1. Introduction

1.1 Problem Summary and Introduction

In many urban areas, residents often encounter difficulties when attempting to report everyday civic problems such as potholes, uncollected garbage, leaking water pipelines, or malfunctioning streetlights[1]. Traditionally, these issues are reported through methods like phone calls, written applications, or in-person visits to municipal offices[1].

These manual procedures are usually time-consuming and can result in communication gaps, slow responses, and limited transparency. Citizens frequently remain uncertain about whether their complaint has been registered or how long it will take to be resolved[1]. On the administrative side, managing and prioritizing complaints becomes challenging because the information is often scattered and poorly organized[6].
The fundamental problem lies in the lack of a straightforward and dependable digital platform that enables direct interaction between citizens and municipal authorities[1],[2]. Without proper tracking systems and structured data management, many reported problems remain unresolved for extended periods[6]. This situation not only impacts public safety but also lowers the standard of urban infrastructure and overall citizen satisfaction[1].

To overcome these challenges, the proposed project presents a crowdsourced digital solution designed to simplify the process of reporting civic issues online[6].

Through this system, users can submit complaints by providing details such as descriptions, locations, and photographs, and they can monitor the status of their requests in real time. The platform follows a structured process that includes complaint submission, verification, task assignment, and resolution[3]. Separate dashboards for citizens, administrators, and field staff enhance transparency, accelerate response times, and improve communication among all involved parties[2].

1.2 Aim and Objectives of the Project

The primary aim of this project is to develop a simple, dependable, and web-based platform that enables citizens to report common civic problems easily. In many areas, people encounter issues such as potholes, overflowing garbage, water leaks, and damaged streetlights, yet the process of reporting these problems is often slow and confusing[1]. This project seeks to simplify the reporting procedure by offering a digital system where citizens can submit complaints and receive updates about their status[1].
The system is designed to allow users to report issues by entering essential information, including a description of the problem, its location, and supporting images if available. It also enables citizens to monitor the progress of their complaints, giving them assurance that their concerns are being addressed[2],[3].
From the administrative perspective, the system assists authorities by collecting complaints in a centralized database, enabling proper task assignment, and allowing status updates as work progresses. By keeping records organized and accessible, the platform improves transparency, minimizes delays, and helps authorities respond more effectively. Overall, the project encourages public participation in civic improvement and supports better urban management through the use of technology[5].

1.2.1 System Usability and Accessibility

A key objective of this project is to ensure that the system is user-friendly and accessible to a wide range of users, including those with limited technical knowledge. Many individuals hesitate to report civic problems because existing reporting methods are complicated or difficult to understand. To address this, the platform is designed with a clear and simple interface[1],[6].

The system includes essential features such as user registration, login, issue reporting, and complaint tracking, all designed to be easy to navigate. Citizens can quickly submit complaints by filling in basic details and, if necessary, attaching images or selecting a location[3]. By focusing on simplicity and accessibility, the system encourages greater public participation in reporting civic issues, allowing authorities to respond more efficiently.

1.2.2 Transparency and Accountability

Another important objective of the project is to increase transparency and accountability in the process of resolving civic problems. In many traditional reporting systems, citizens submit complaints but rarely receive updates, which leads to dissatisfaction and lack of trust[1],[2].

This system allows users to track their complaints at different stages, such as pending, in progress, or resolved. Administrators and staff are responsible for updating the status as actions are taken, ensuring that each step in the process is recorded and visible[4]. This approach makes the process more transparent and helps ensure that responsible departments complete their tasks on time.

1.2.3 Data Organization and Analysis

Proper management of complaint data is another major objective of this project. In traditional or poorly organized systems, records are often scattered, duplicated, or lost, which causes delays and confusion. This system stores all complaint information, user details, location data, and status updates in a structured and centralized database.

Organized data allows authorities to identify frequently occurring problems and recognize areas where issues are more common. Such insights can support better planning and decision-making in the future[5]. Effective data management also reduces duplicate complaints and improves coordination among departments, leading to more efficient civic administration.

1.3 Problem Specifications

In many cities, reporting civic issues remains a slow and inconvenient process. Citizens often rely on phone calls, personal visits to offices, or informal channels to report problems such as damaged roads, garbage accumulation, or water leakage. These methods are usually unstructured and may result in delays, lost complaints, or inaccurate information. The absence of a unified reporting platform creates confusion for both citizens and authorities[1],[3].

Another significant challenge is the lack of complaint tracking. After submitting a report, citizens often receive no information about the progress or resolution of their issue. This uncertainty reduces trust in the system. At the same time, authorities struggle to manage and prioritize complaints because reports come from multiple sources and are not stored in a systematic manner[5].

Poor data handling further complicates the situation. Complaints may be duplicated, misplaced, or delayed due to inefficient record keeping[3]. In addition, unclear roles and responsibilities among administrative staff can reduce accountability and slow down resolution. These challenges highlight the need for a structured, transparent, and digital solution that simplifies reporting, improves data management, and ensures timely resolution

of civic problems[2],[6].

1.4 Literature Review

Recent research in the field of crowdsourced civic reporting and digital governance indicates a growing adoption of web-based platforms and data-driven systems to improve transparency, response time, and citizen engagement[1]. Over the past decade, many cities have moved from manual complaint handling methods to centralized digital platforms that support real-time reporting and tracking. Studies show that such systems help reduce delays, strengthen communication between citizens and authorities, and improve civic services[1]. Several researchers have identified transparency and traceability as critical factors for the success of civic reporting platforms. Systems that provide unique complaint IDs, real-time updates, and clear workflows for task assignment tend to achieve higher levels of public participation and trust[4]. Evidence suggests that when citizens can monitor the progress of their complaints, they are more likely to engage actively and provide accurate information. Recent studies also emphasize the importance of effective task allocation and workload management in administrative and distributed systems. Proper assignment of responsibilities and organized tracking of tasks can significantly improve the efficiency of issue resolution and overall system performance[6].

Table 1.1 Literature Review Summary Table

Ref No.	Topic Name	Paper Title	Year	Objective	Key Contribution
[1]	Crowdsourced civic issue reporting platforms	SeeClickFix: Distributed Democracy for Crowdsourced Issue Reporting – Mergel, I.	2012	To study how neighborhood reporting platforms enable citizen reporting and official response	Transparent issue tracking increases trust; assignment and status updates improve citizen engagement.
[2]	Blockchain for transparent civic reporting	Blockchain as Enabling Technology for Transparent Civic Reporting – Farnaghi, M.	2020	To examine how blockchain enhances transparency and tamper-resistance.	Tamper-proof audit trails improve accountability in reporting systems.
[3]	Task allocation in distributed systems	Task Allocation in Distributed Systems – Patsis, V., et al. (MDPI)	2023	To optimize task assignment across distributed agents.	Hybrid assignment methods reduce latency and balance system load.
[4]	Anonymous authentication in crowdsensing	Efficient Anonymous Authentication and Privacy-Preserving Reliability Evaluation for Mobile Crowdsensing in Vehicular Networks – Cheng, J., et al.	2023	To enable credible yet anonymous participation in mobile crowdsensing.	Cryptographic pseudonymous authentication maintains anonymity while preserving credibility.
[5]	Privacy risks of national ID authentication	Privacy and Security of Aadhaar – Agrawal, S., et al	2015	To analyze risks of using national ID systems for authentication.	Identifies privacy concerns and potential misuse risks in Aadhaar-based systems.
[6]	Automated triage for incident management	AIOps and Automated Triage for Incident Management – Gao, X., et al.	2024	To reduce human workload through ML-based automated triage.	Automated triage improves resolution efficiency and prioritization of incidents.

1.5 Plan of the Work

The work for this civic issue reporting and resolution project is planned and executed in a structured and systematic manner to ensure smooth development and effective implementation. The project is carried out in multiple stages, where each stage focuses on understanding the problem, designing the system, developing the required modules, and testing the overall functionality. This planned approach helps in building a reliable system that meets both citizen and administrative requirements[1],[4].

1.5.1 Requirement Analysis

The requirement analysis phase focuses on understanding the real-world problems faced by citizens while reporting civic issues and the challenges faced by municipal authorities in managing these complaints. During this stage, common civic problems such as potholes, garbage overflow, water leakage, and damaged streetlights are studied along with the limitations of existing manual reporting systems[6]. Functional requirements such as user registration, issue reporting with location and images, complaint tracking, and role-based access for administrators and staff are identified. Non-functional requirements including system usability, security, performance, and scalability are also analyzed to ensure that the system is reliable, efficient, and suitable for real-world usage[1].

1.5.2 System Design

The system design phase involves planning the overall structure and workflow of the civic issue reporting system. This includes designing the architecture that connects the frontend interface, backend logic, and database. The data flow is structured from the point where a citizen submits an issue to the stages of verification, assignment, progress update, and final resolution. Separate dashboards are designed for users, administrators, and staff to ensure clarity in responsibilities and smooth coordination. Emphasis is placed on creating a simple yet effective design that supports transparency and efficient issue management[5].

1.5.3 Implementation

In the implementation phase, the designed system is developed using appropriate web technologies. Features such as user authentication, civic issue submission, image upload, and location handling are implemented to support citizen interaction. Admin and staff modules are developed to allow verification of complaints, assignment of tasks, and updating of issue status[2],[4]. The database is implemented to store user details, issue records, and resolution updates in an organized manner. This phase focuses on converting system design into a functional and working application[1].

1.5.4 Testing and Validation

The testing and validation phase ensures that the system functions correctly and meets all defined requirements. Each module of the system is tested individually to verify features such as login, issue reporting, status tracking, and role-based access control. Integration testing is also performed to ensure smooth interaction between the frontend, backend, and database. Usability testing is carried out to confirm that the system is easy to use for citizens and authorities[1],[2]. Validation ensures that the system produces accurate and reliable results under different usage scenarios[1].

1.5.5 Deployment and Documentation

The final phase of the project involves deployment and documentation. The system is deployed on a local or web server environment to make it accessible for users and administrators. Proper configuration of the server and database is carried out to ensure stable performance. Along with deployment, detailed documentation is prepared to explain system features, user roles, workflows, and technical implementation[1]. This documentation helps future users, administrators, and developers understand, operate, and maintain the system effectively. The project is then finalized with provisions for future upgrades and enhancements[1].

1.6 Materials / Tools Required

The development and implementation of the Crowdsourced Civic Issue Reporting and Resolution System require a set of reliable and commonly used software tools. These tools help in building, testing, and running the web-based application smoothly. All the tools selected for this project are easy to use, widely accepted in the industry, and suitable for developing a secure and scalable civic issue reporting system[1].

The frontend of the system is developed using basic web technologies that allow users to interact with the application easily. These tools help in creating a clean and user-friendly interface where citizens can register, log in, report issues, and track their complaint status.

• HTML – Used to design the basic structure of web pages such as login, registration.
• CSS – Used to improve the appearance of the application and make the interface visually attractive.
• JavaScript – Used for client-side validation and to enhance user interaction within the web application.

The backend of the system is responsible for processing data and managing the application logic. PHP is used as the server-side language to handle important system operations securely and efficiently.

• PHP – Used for server-side processing, user authentication, session management, issue submission handling, image uploads, and role-based access control for users, administrators, and staff.

For storing and managing all civic issue-related data, a reliable database system is required. MySQL is used to store structured data such as user details, issue records, locations, images, and status updates.

• MySQL – Used for database management to store, retrieve, and manage civic issue data in an organized and secure manner.

To run and test the application during development, a local server environment is required. This allows developers to test the system before deploying it online.

• XAMPP / WAMP – Used as a local server environment that provides Apache server.
• Web Browser (Google Chrome / Microsoft Edge) – Used to access, test, and verify the functionality of the web application.

2. Analysis, Design Methodology and Implementation Strategy

2.1 System Overview

The Crowdsourced Civic Issue Reporting and Resolution System is a web-based application designed to make the process of reporting and resolving civic issues simple, transparent, and efficient[3]. The system provides a digital platform where citizens can report common problems such as potholes, garbage overflow, water leakage, broken streetlights, and other public infrastructure issues. Instead of relying on manual methods like phone calls or office visits, users can submit complaints online along with relevant details such as descriptions, images, and location information[4].

The system is designed with different user roles to ensure smooth and organized operation. Citizens can register, log in, report issues, and track the progress of their complaints through a user-friendly dashboard[5]. Administrators have access to a centralized dashboard where they can view all reported issues, verify complaints, assign tasks to staff members, and monitor overall system activity[1]. Staff members are responsible for handling assigned issues and updating the status as work progresses. This role-based structure helps maintain accountability and improves coordination among all stakeholders.

The backend of the system handles data processing, authentication, and communication with the database, ensuring secure storage and management of information. All civic issue data is stored in a centralized database, which helps in avoiding duplication, maintaining consistency, and enabling easy analysis of recurring problems. Overall, the system aims to improve civic service delivery by encouraging citizen participation, ensuring transparency, and supporting faster and more effective issue resolution[6].

2.2 Observation Matrix

The observation matrix is used to study the existing civic issue reporting system and identify its limitations[2]. This analysis helps in understanding the real problems faced by citizens and municipal authorities while handling civic complaints. Based on these observations, suitable improvements are proposed in the form of a digital and crowdsourced solution[2]. The comparison highlights how the proposed system overcomes the drawbacks of the existing system and improves efficiency, transparency, and user experience.

Table 2.1 Observation and Analysis Table

Observation Domain	Key Issue Identified	Effect on Citizens / Authorities	Derived Insight
Issue Reporting Process	Civic issues are reported using manual methods such as phone calls or office visits	Citizens face delays and inconvenience, while authorities receive incomplete or unclear complaints	A digital reporting platform is required to simplify and standardize issue submission
Complaint Tracking	No proper mechanism to track the status of reported issues	Citizens remain unaware of complaint progress, leading to frustration and loss of trust	Real-time complaint tracking improves transparency and user confidence
Transparency	Lack of visibility in how complaints are handled	Citizens feel ignored and authorities are not held accountable	Transparent status updates help build trust and accountability
Data Management	Complaint data is scattered and poorly organized	Authorities struggle to manage, analyze, and prioritize issues	Centralized data storage improves organization and decision-making
Communication Flow	Limited communication between citizens and civic authorities	Miscommunication causes delays in issue resolution	Digital dashboards enable better coordination and faster response
Accountability	No clear responsibility assigned for complaint resolution	Difficulty in tracking who is responsible for resolving issues	Role-based access ensures clear ownership and responsibility
User Experience	Reporting civic issues requires repeated follow-ups	Low participation due to inconvenience	A user-friendly system encourages active citizen involvement

2.3 Technology Stack

The technology stack selected for this project plays an important role in building a reliable, efficient, and user-friendly Crowdsourced Civic Issue Reporting and Resolution System. The chosen technologies are widely used in web application development and are well suited for handling real-time data, user interaction, and secure data storage[1].

For the frontend, **HTML, CSS, and JavaScript** are used to design and develop the user interface of the system. HTML is responsible for structuring the web pages such as registration, login, dashboards, and issue reporting forms. CSS is used to improve the visual appearance of the application by making it responsive, clean, and easy to navigate. JavaScript enhances the user experience by handling client-side validation and improving interactivity within the

system[2].

The backend of the application is developed using **PHP**, which manages all server-side operations. PHP handles user authentication, complaint submission, image uploads, session management, and role-based access for citizens, administrators, and staff members[1]. It also acts as a bridge between the frontend and the database, ensuring smooth and secure data processing.

For data storage and management, **MySQL** is used as the database. It stores all important information such as user details, civic issue records, locations, images, and complaint status updates in a structured manner. MySQL ensures data consistency, fast retrieval, and reliable handling of large volumes of records[1],[2].

The system is developed and tested using a local server environment such as **XAMPP or WAMP**, which provides Apache server support along with PHP and MySQL. A standard web browser like **Google Chrome or Microsoft Edge** is used for testing and accessing the application during development[4].

Overall, this technology stack provides a balanced combination of simplicity, performance, and scalability, making it suitable for developing an effective civic issue reporting system[5].

Table 2.2 Technology Stack of the System

Component	**Technology Used**	**Purpose**
Frontend Interface	Web-based dashboard using HTML, CSS, and JavaScript	To provide a simple, responsive, and user-friendly interface for citizens to report civic issues and track their status
Backend Server	PHP	To handle server-side operations such as user authentication, complaint submission, role-based access, and system logic
Database	MySQL	To store and manage user details, civic issue records, images, locations, and complaint status updates in an organized manner
Web Server	Apache (via XAMPP/WAMP)	To host and run the web application locally during development and testing
Development Environment	XAMPP / WAMP	To provide an integrated environment with PHP, MySQL, and Apache for smooth development
Client Access	Web Browser (Google Chrome / Microsoft Edge)	To access, test, and interact with the civic issue reporting system
Image Handling	File upload module in PHP	To allow users to upload images as evidence.
Security & Access Control	Session management and role-based login	To ensure secure access for citizens, administrators, and staff members

2.4 Database Design

The database design of the Crowdsourced Civic Issue Reporting and Resolution System plays a crucial role in storing, managing, and retrieving civic issue data efficiently. A well-structured database is essential to ensure data consistency, reduce redundancy, and support smooth system operations. The database is designed using MySQL and follows a relational model to organize information in a clear and structured manner[6].

The database consists of multiple tables that store different types of information related to users, civic issues, and system activities. User-related data such as user ID, name, contact details, role, and login credentials are stored in a dedicated user table. This table helps in identifying different users and enforcing role-based access within the system.

Civic issue details are stored in a separate table that includes information such as issue ID, issue category, description, location, image path, submission date, and current status[3]. This table forms the core of the database and allows authorities to track each complaint from submission to resolution[1],[4].

To support issue handling and accountability, an assignment table is used to map civic issues to staff members. This table stores assignment details such as staff ID, issue ID, and assigned date, ensuring that responsibilities are clearly recorded. Additionally, a status history table is maintained to track all updates made to an issue over time, providing transparency and complete complaint history[4].

The database also includes tables for storing uploaded images and administrative actions, ensuring proper linkage between issues and supporting evidence. Relationships between tables are established using primary and foreign keys, which helps maintain data integrity and enables efficient querying.

Overall, the database design supports reliable data storage, quick retrieval of information, and future scalability[1]. It enables effective reporting, analysis of recurring civic problems, and long-term record maintenance, making it a strong foundation for the proposed system.

2.5 System Design

The system design of the Crowdsourced Civic Issue Reporting and Resolution System focuses on creating a simple, efficient, and reliable platform that connects citizens with civic authorities. The design ensures that all components of the system work together smoothly while maintaining ease of use, transparency, and proper data management[3].

The system is designed as a web-based application that can be accessed through a standard web browser. It follows a modular design approach, where the frontend, backend, and database are developed as separate but interconnected components[2]. This separation makes the system easier to maintain, test, and enhance in the future.

The frontend design provides different interfaces for citizens, administrators, and staff members based on their roles. Citizens are given a user-friendly interface to register, log in, report civic issues, upload images, and track complaint status. The interface is designed to be simple and intuitive so that users from different age groups and technical backgrounds can use the system without difficulty[6].

The backend design handles all core system logic such as user authentication, issue submission, validation, role-based access control, and status updates. It ensures secure communication between the user interface and the database. The backend also manages complaint assignment,

staff coordination, and administrative actions, ensuring smooth system operations[1].

The database design supports structured storage of user information, civic issue details, images, assignments, and status history. Proper relationships between tables are maintained to ensure data integrity and efficient retrieval. Centralized data storage allows easy reporting, analysis, and long-term record maintenance.

Overall, the system design ensures reliability, scalability, and transparency. It supports efficient civic issue handling while providing a positive user experience for citizens and effective management tools for authorities[2],[3].

2.5.1 Use Case Diagram

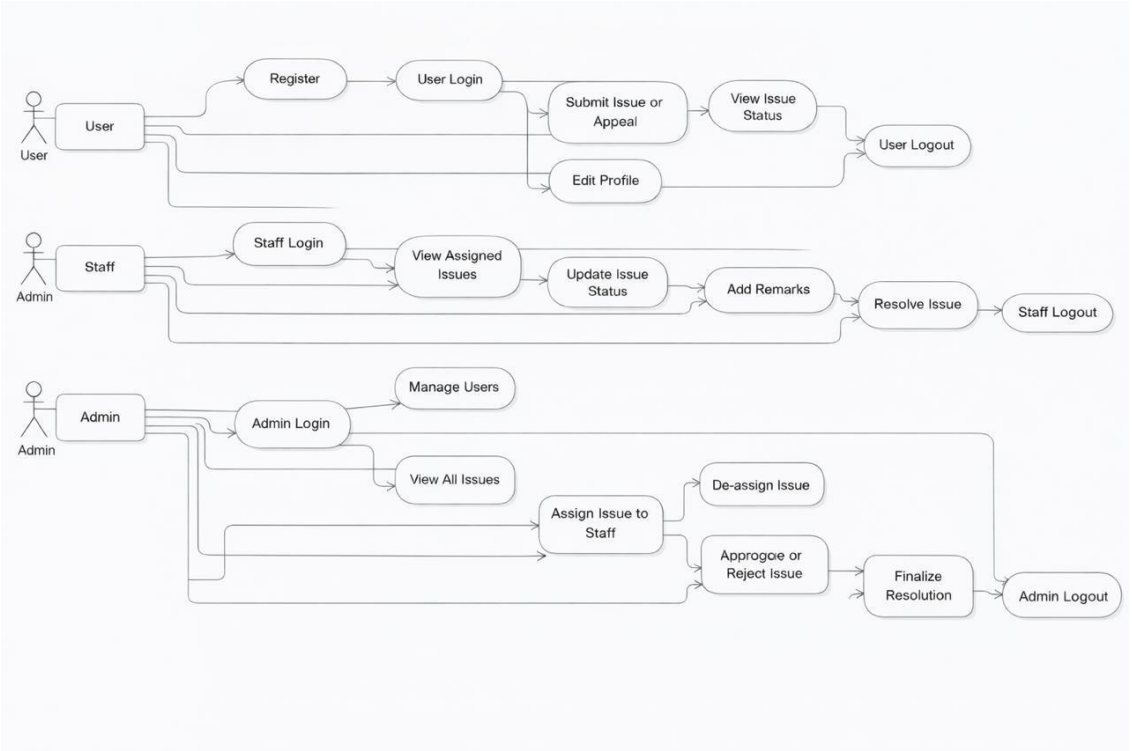

Fig 2(a) Use Case Diagram of Croweded Civic Issue Reporting and Resolution System

2.5.2 Class Diagram

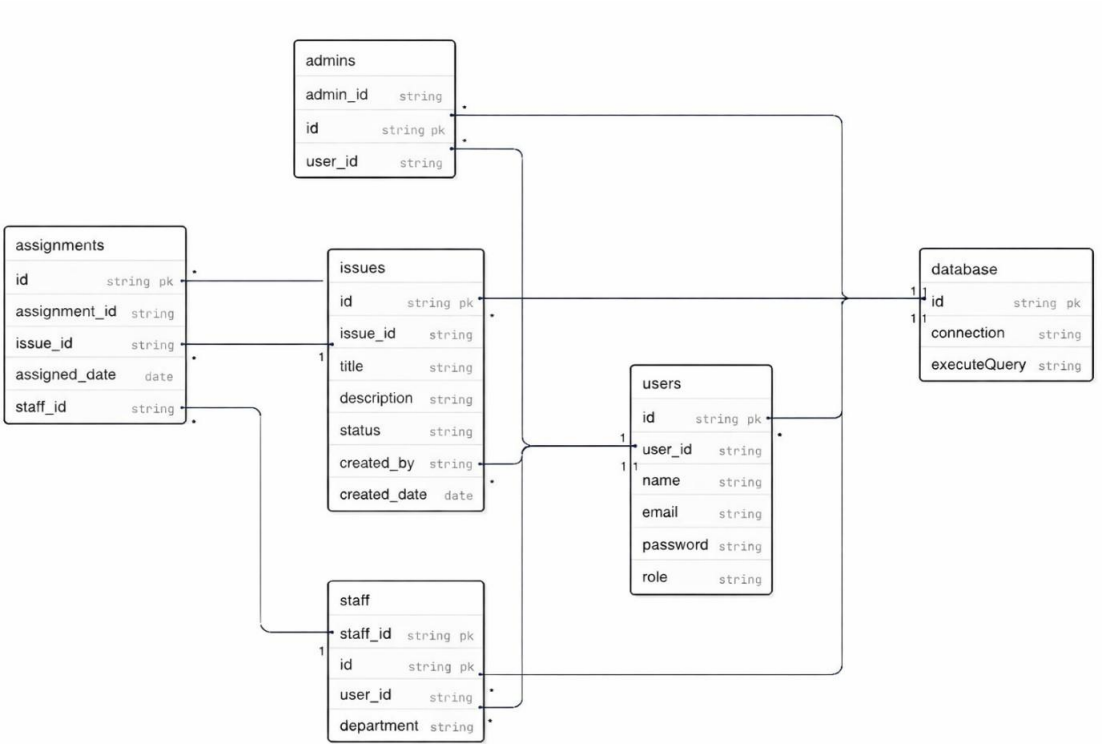

Fig 2(b) Class Diagram of Croweded Civic Issue Reporting and Resolution System

2.5.3 Activity Diagram

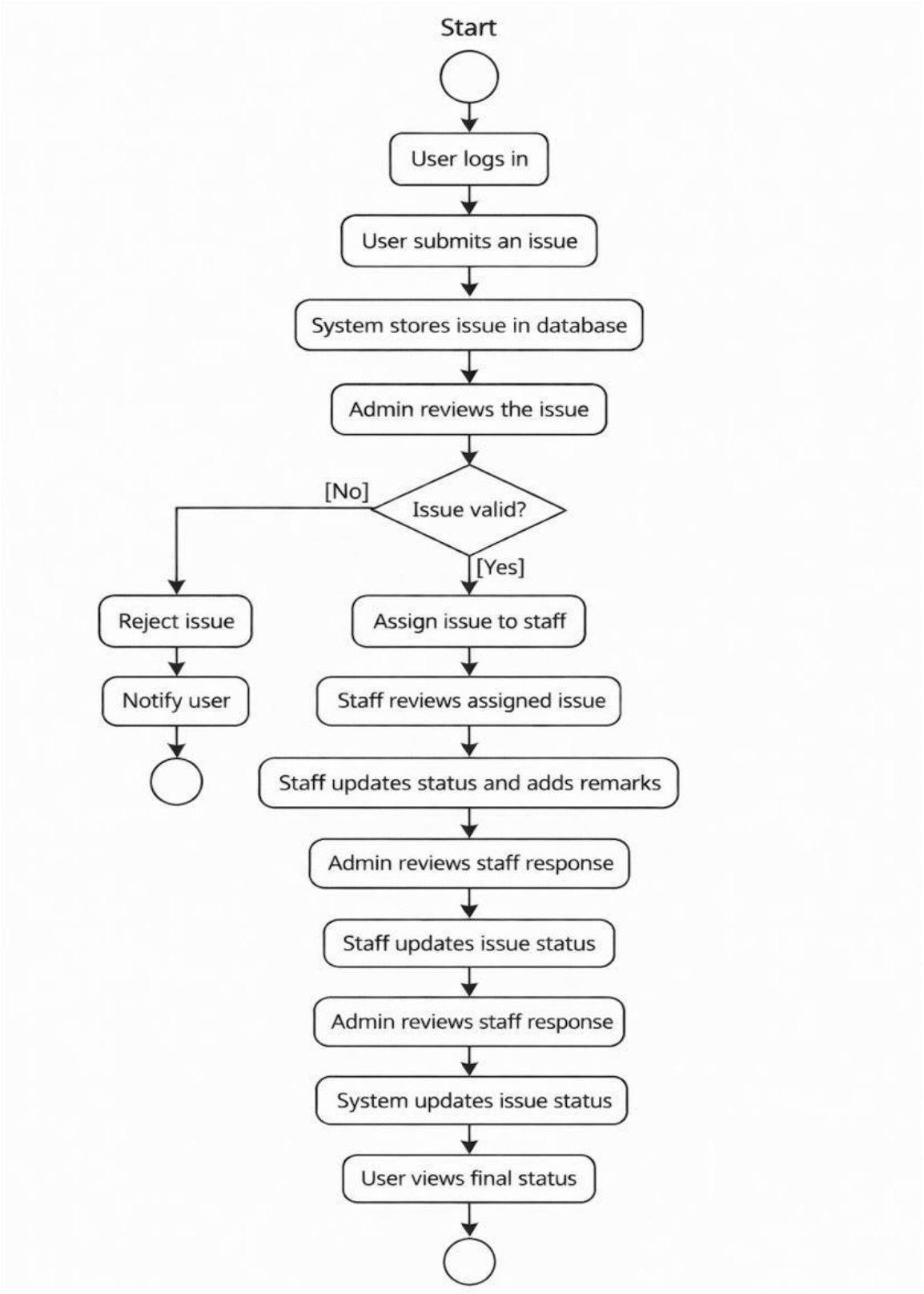

Fig 2(c) Activity Diagram of Croweded Civic Issue Reporting and Resolution System

2.5.4 Sequence Diagram

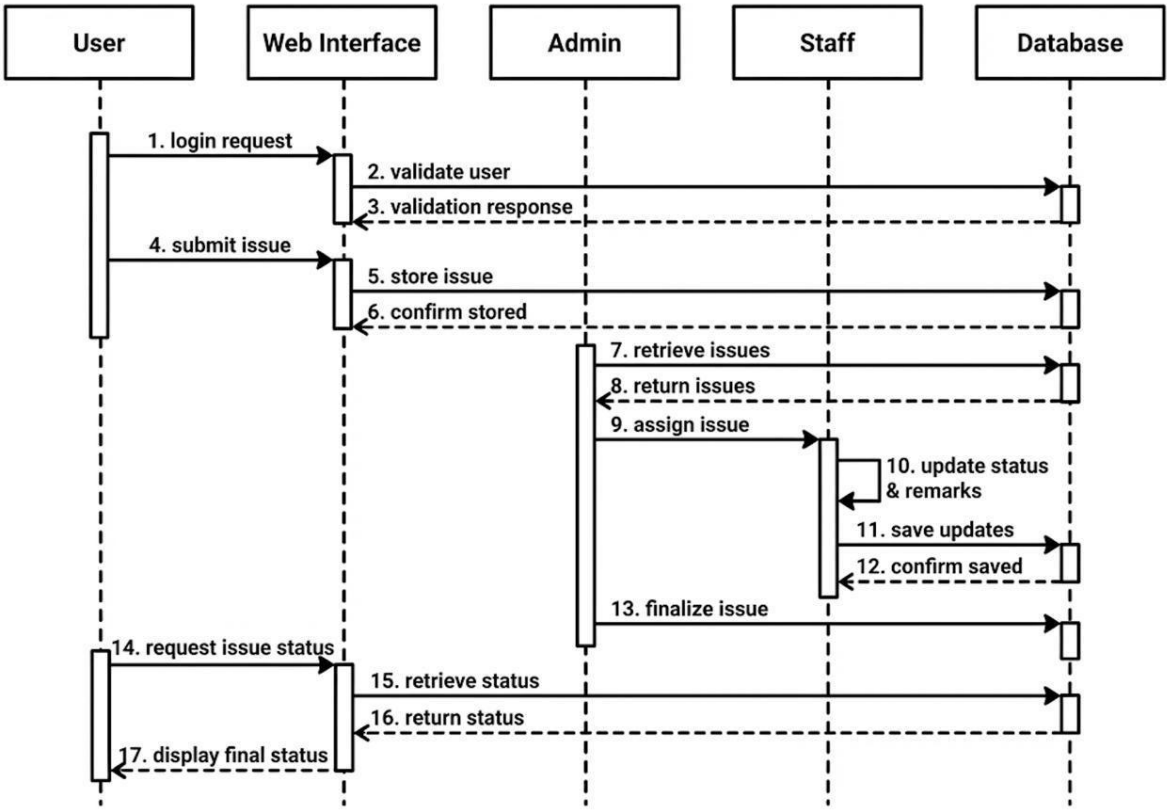

Fig 2(d) Sequence Diagram of Croweded Civic Issue Reporting and Resolution System

2.6 System Architecture

The system architecture of the Crowdsourced Civic Issue Reporting and Resolution System is designed to support smooth communication between users, the application logic, and the database. The architecture follows a web-based client–server model, which helps in organizing system components and improving scalability and reliability[3].

At the client side, users interact with the system through a web browser. Citizens, administrators, and staff members access the application using dashboards designed according to their roles. The frontend is built using HTML, CSS, and JavaScript, which ensures a responsive and user-friendly interface for reporting issues, tracking complaint status, and managing civic tasks[4].

The application layer acts as the core of the system and is developed using PHP. This layer handles all server-side processing such as user authentication, complaint submission, image uploads, issue verification, role-based access control, and status updates. It also manages communication between the frontend and the database while ensuring secure data handling[2].

The database layer uses MySQL to store all system-related data in a structured manner. This includes user information, civic issue details, location data, uploaded images, and complaint status history. The database ensures data consistency, quick retrieval, and reliable storage of records[1].

All components of the system are hosted on an Apache web server, typically through a local

development environment such as XAMPP or WAMP. This architecture ensures smooth system performance, easy maintenance, and the ability to scale the system in the future if required[1].

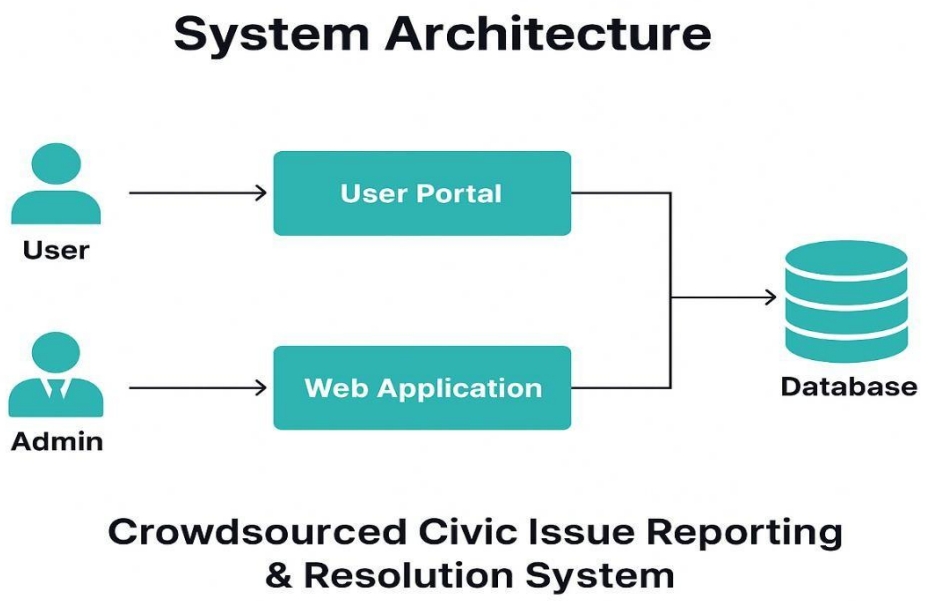

Fig 2(e) System Architecture Diagram

2.6.1 User Module Architecture

The user module forms the entry point of the system where citizens interact with the application through the user portal. This module allows users to register, log in, report civic issues, upload images, and track the status of their complaints[5]. The design of this module focuses on simplicity and ease of use so that people from different age groups and technical backgrounds can report issues without difficulty. All user actions are securely forwarded to the web application layer for processing[3].

2.6.2 Admin Module Architecture

The admin module is responsible for managing and monitoring all reported civic issues. Through the web application interface, administrators can view incoming complaints, verify their authenticity, assign issues to staff members, and monitor progress[2]. This module ensures proper coordination between departments and helps maintain accountability in the issue resolution process. The admin module plays a crucial role in maintaining system control and workflow efficiency[2].

2.6.3 Web Application Layer

The web application layer acts as the core processing unit of the system. It handles business logic such as authentication, issue validation, role-based access control, and status updates. This layer connects both the user portal and admin interface to the database, ensuring secure and smooth data exchange. By separating the application logic from the user interface, the system becomes easier to maintain and scale in the future[1].

2.6.4 Data Flow Mechanism

The data flow in the system follows a clear and structured path. User inputs such as complaint details and images are submitted through the user portal and processed by the web application layer before being stored in the database[1]. Any updates made by administrators or staff are reflected back to the user through real-time status updates. This smooth data flow ensures transparency and reduces delays in issue resolution[1].

2.6.5 Security and Access Control

Security is an important aspect of the system architecture. Role-based access control ensures that citizens, administrators, and staff members can access only the features relevant to their roles. Session management and authentication mechanisms help protect user data and prevent unauthorized access. This approach ensures data privacy and system reliability[4].

3. Implementation

3.1 Implemented Functionality

The Crowdsourced Civic Issue Reporting and Resolution System is developed with several key functionalities that allow citizens, administrators, and staff members to interact effectively with the platform. Each functionality is designed to make the process of reporting and resolving civic issues simple, transparent, and efficient[1].

The system provides a **user registration and login** feature that allows citizens to create an account and securely access the application. Once logged in, users can submit civic issues by selecting the issue category, entering a description, uploading supporting images, and specifying the location of the problem. This functionality ensures that complaints are recorded in a structured and clear manner[2],[5].

Another important functionality is **complaint tracking**, which allows users to view the current status of their reported issues. Status updates such as “pending,” “in progress,” and “resolved” help citizens stay informed about the progress of their complaints, improving transparency and trust[6].

The system also includes an **admin management module**, where administrators can review submitted complaints, verify their validity, and assign them to appropriate staff members. This functionality helps in proper workload distribution and faster issue resolution.

For staff members, the system provides a **task handling module** that displays assigned complaints along with necessary details. Staff members can update the status of issues after taking action, ensuring real-time updates for both administrators and citizens[3].

Additionally, the system implements **role-based access control**, ensuring that users, administrators, and staff members can access only the features relevant to their roles. All data related to users and complaints is stored securely in the database, ensuring reliability and future reference.

Overall, the implemented functionalities work together to provide an efficient, user-friendly, and transparent civic issue reporting and resolution platform[1].

3.1.1 User Registration and Authentication

The system provides a secure and reliable user registration and authentication mechanism for all types of users, including citizens, administrators, and staff members. During registration, users are required to provide basic personal details, which helps in maintaining the authenticity of complaints and reducing misuse of the system. The login functionality ensures that only authorized users can access the system, while session management prevents unauthorized access. This feature not only enhances system security but also builds trust among users by ensuring that every reported issue is linked to a genuine and identifiable account[1].

3.1.2 Civic Issue Reporting

Civic issue reporting is the core functionality of the system, designed to make the complaint submission process simple, clear, and effective. Citizens can report issues by selecting the appropriate category, providing a detailed description of the problem, uploading supporting images, and specifying the exact location. This structured approach ensures that complaints are well-documented and easy for authorities to understand. By allowing users to submit issues online at any time, the system removes the dependency on manual reporting methods and

encourages greater public participation[1],[2].

3.1.3 Issue Status Tracking

Once a civic issue is reported, the system enables users to track the progress of their complaint through a dedicated dashboard. The complaint status is updated at various stages such as pending, under review, in progress, and resolved. These real-time updates help citizens stay informed about the actions being taken without the need to repeatedly contact authorities. This functionality plays a significant role in improving transparency and reducing frustration caused by uncertainty and delays[6].

3.1.4 Admin Dashboard and Issue Management

The admin dashboard serves as the central control unit of the system, providing administrators with complete visibility of all reported civic issues. Administrators can review complaints, verify their validity, assign them to appropriate staff members, and monitor progress. The dashboard allows administrators to manage multiple issues efficiently, prioritize urgent complaints, and ensure that no issue is overlooked[5]. This centralized management functionality greatly improves coordination and accountability within the system.

3.1.5 Staff Task Management

Staff members are provided with a dedicated interface that displays all complaints assigned to them. Each complaint includes detailed descriptions, images, and location information to help staff members understand the issue clearly. After taking necessary action, staff members can update the complaint status and add remarks regarding the resolution. This functionality ensures smooth communication between staff, administrators, and citizens, while also enabling faster and more organized issue resolution[1].

3.1.6 Image Upload and Evidence Handling

The system allows users to upload images while reporting civic issues, which act as visual evidence of the problem. These images help authorities assess the severity and authenticity of complaints more accurately. The image upload feature also reduces false or vague reporting, as users are encouraged to provide clear proof of the issue. This functionality supports better decision-making and helps prioritize issues that require immediate attention[2],[5].

3.2 Results and Reports

The results of the Crowdsourced Civic Issue Reporting and Resolution System demonstrate that the application successfully achieves its intended objectives of improving civic issue reporting, transparency, and communication between citizens and authorities[1]. After implementation and testing, the system was able to efficiently handle user registrations, issue submissions, complaint tracking, and administrative operations without major errors.

The system results show that citizens are able to report civic issues easily through the web-based interface. Complaints submitted by users are stored accurately in the database along with supporting images and location details. The structured reporting process ensures that issues are clearly documented, making it easier for administrators and staff members to understand and address them effectively[2].

From the administrative perspective, the system provides clear and organized reports of all reported issues. Administrators can view complaint summaries, issue categories, current status, and assigned staff members through the dashboard. These reports help administrators monitor system activity, identify pending and resolved complaints, and take timely action when delays occur[2].

The staff reports generated by the system display assigned tasks, issue details, and status updates. These reports help staff members manage their workload efficiently and ensure that assigned issues are resolved within a reasonable time. The ability to update complaint status reflects immediately in user dashboards, confirming successful communication across the system.

The system also maintains historical reports of previously reported and resolved civic issues. These reports are useful for analyzing recurring problems, identifying high-issue areas, and understanding common civic concerns. Such insights can support better planning and decision-making by municipal authorities in the future[1].

Overall, the results indicate that the system improves efficiency, transparency, and accountability in the civic issue resolution process. The generated reports provide meaningful insights into system performance and user activity, proving the effectiveness of the proposed solution[1].

3.3 Snapshots

Login Page

This screen allows registered users to sign in to the Crowded Source Civic Issues Reporting System to report issues, track progress, and stay informed about civic problem resolutions.

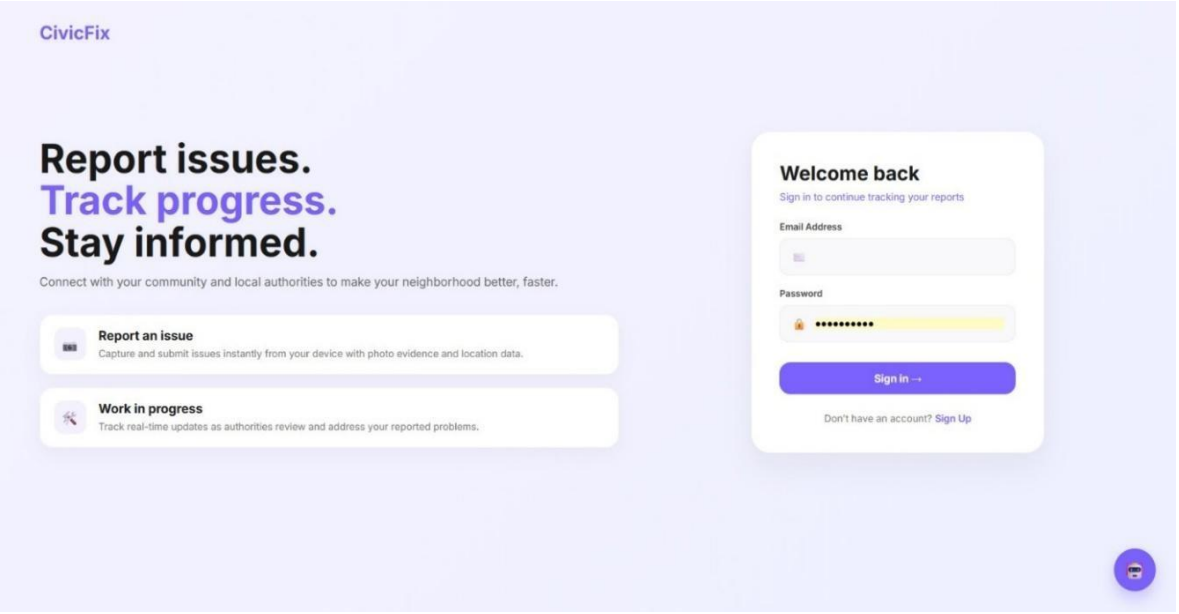

Fig 3(a) Login Page

HOD Dashboard Screen
Shows overall issue statistics, community feed, assignment status, appeals, and resolution overview.

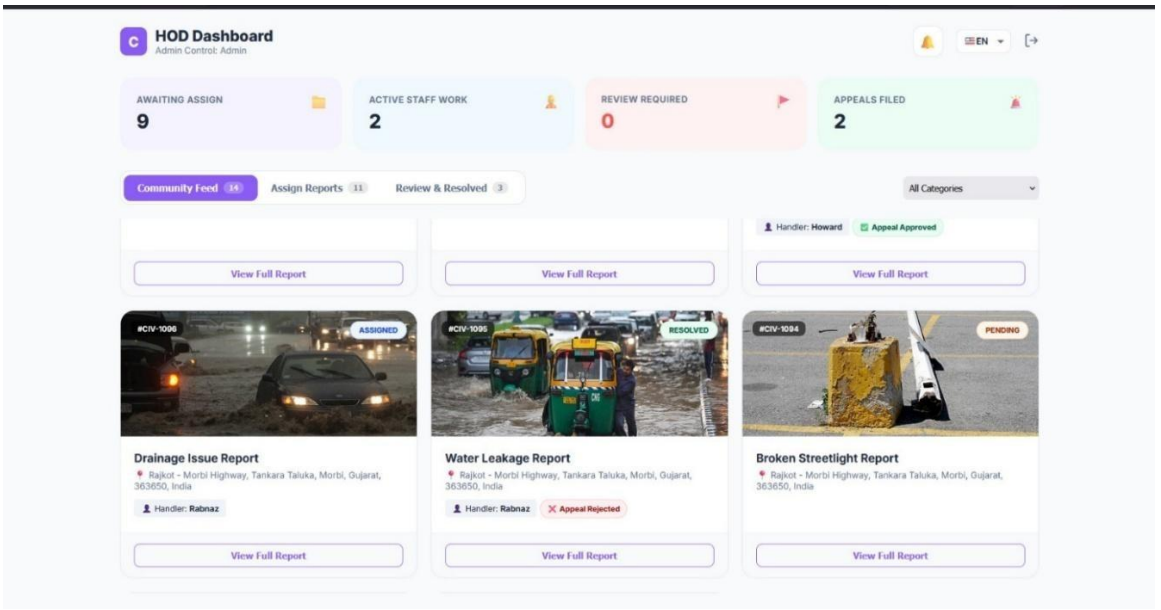

Fig 3(b) HOD Dashboard

Case Details / Full Issue View
Displays complete issue information including description, address, case progress, citizen evidence, and official timeline.

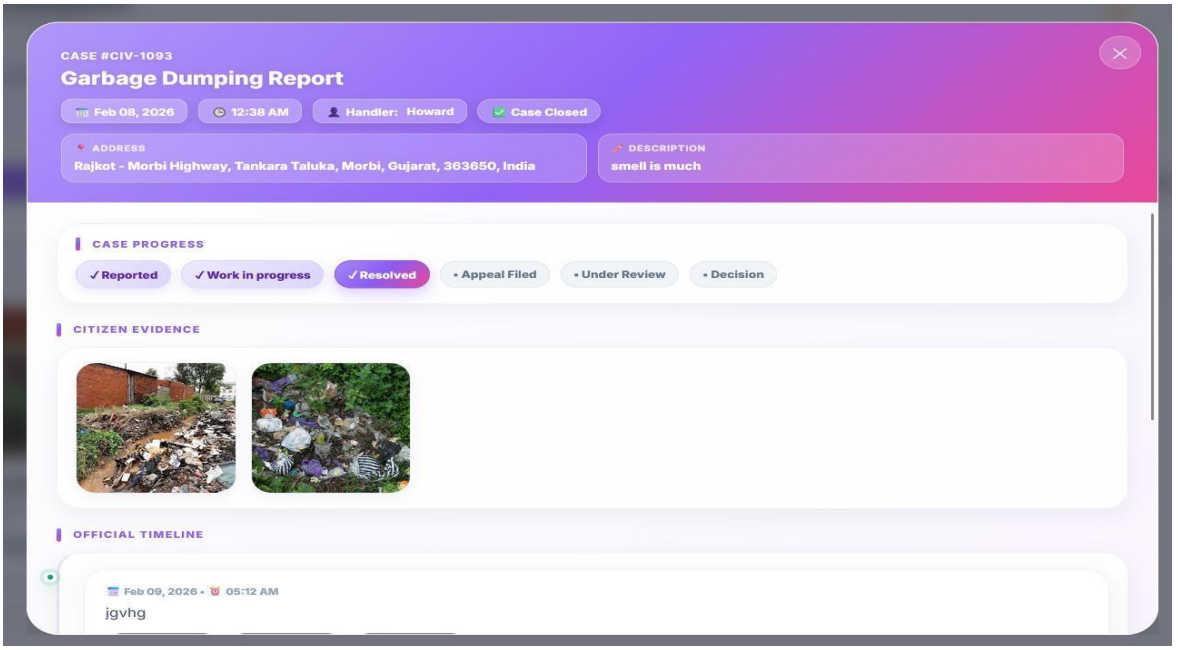

Fig 3(c) Case Details View

Admin Dashboard / Issue Management View
Displays an overview of all civic issues including total issues, assigned tasks, pending cases, and resolved reports. It shows issue cards with category, location, date, images, and current status (pending/resolved). Staff can filter by category or status, monitor progress, and manage community-reported problems efficiently from one dashboard.

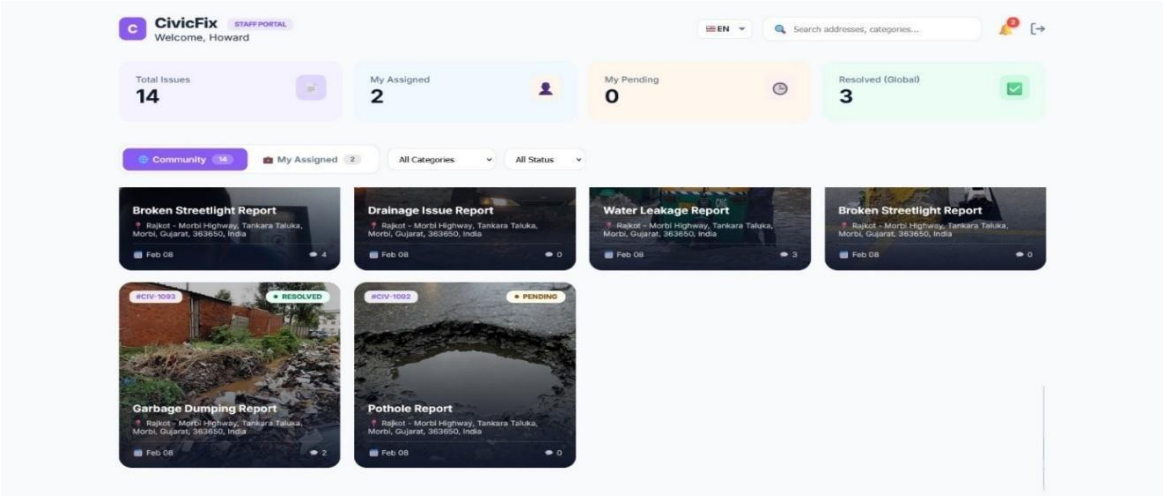

Fig 3(d)Admin Dashboard

Citizen Dashboard / Report Management View
Displays an overview of user-reported civic issues including total reports, pending cases, issues in progress, and resolved complaints. It allows citizens to quickly report new issues, view community reports, track their submitted complaints, filter by category or status, and monitor real-time updates on each issue with images, location, date, and approval status.

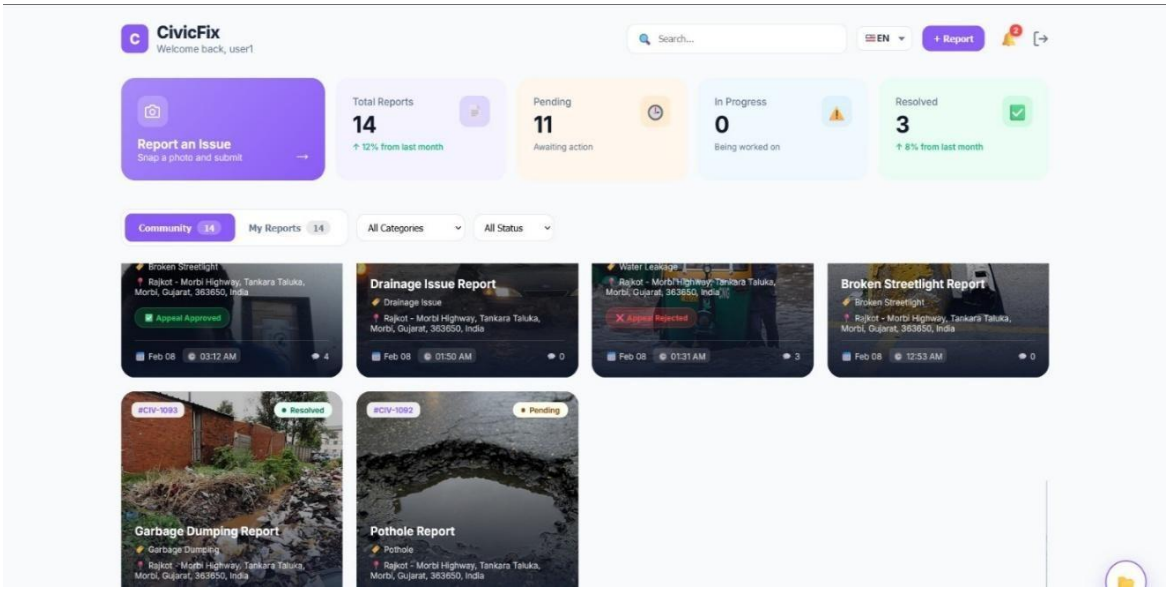

Fig 3(e) User Dashboard

Report Issue / Capture & Submit View

Allows users to report civic problems by capturing or uploading photos, automatically syncing location details, entering address information, and selecting issue categories such as potholes, garbage, water, drainage, or lighting. This screen enables quick and accurate submission of complaints with real-time location and visual evidence.

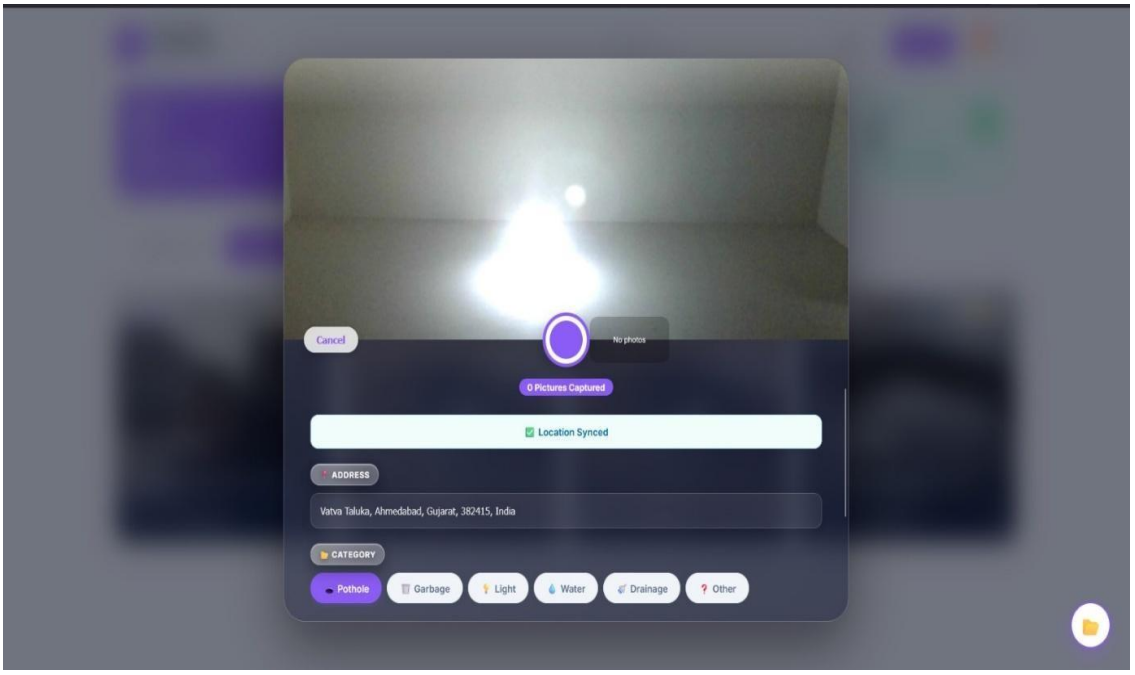

Fig 3(f)Geo-Location Issue Capture

Create Report / Issue Submission Form

Enables users to create and submit a new civic issue report by uploading evidence photos from gallery or library, adding address or GPS location, selecting the issue category, and providing a detailed description. This form ensures accurate and complete reporting of civic problems for quick action by authorities.

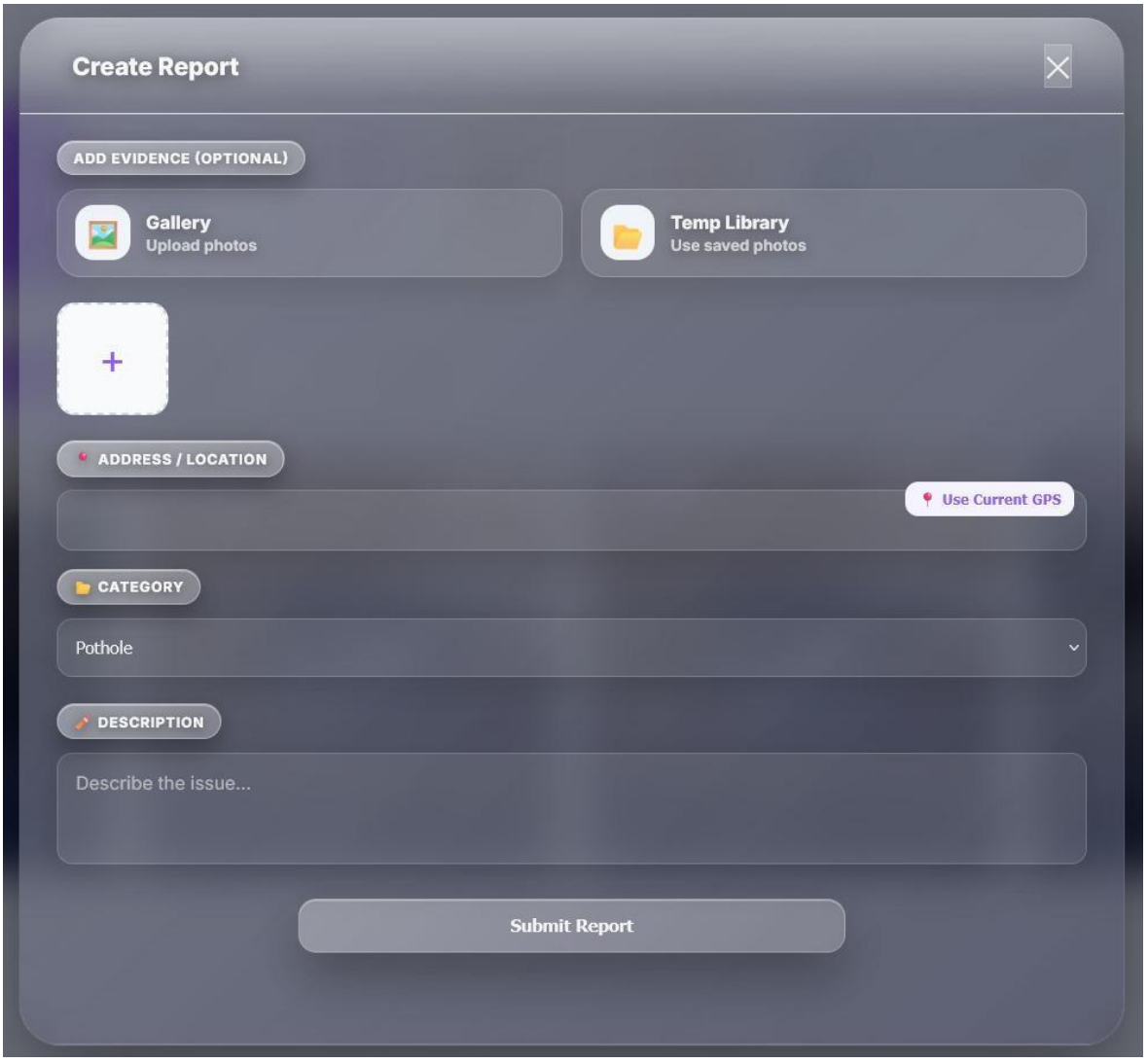

Fig 3(g)Report Submission Form

Temporary Photo Library View

Displays a temporary storage area where users can view and select previously captured or saved images to attach as evidence for a new civic issue report. This feature allows easy access and reuse of photos before submitting the final complaint.

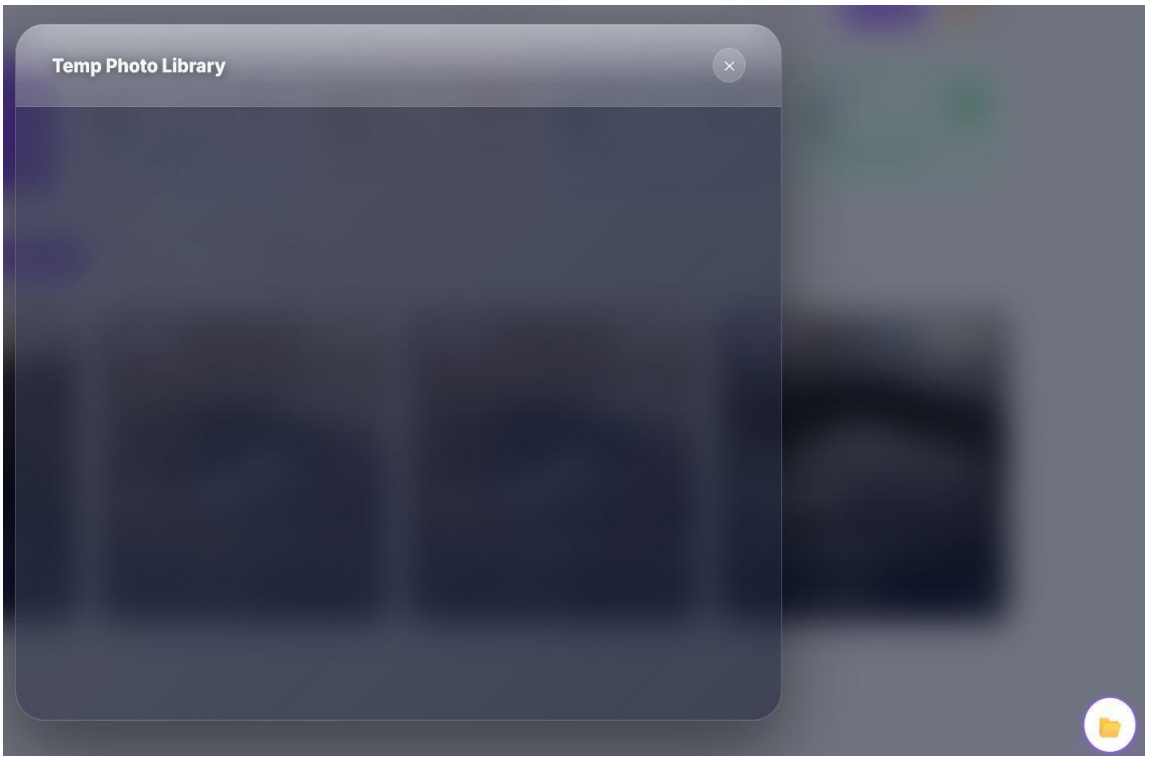

Fig 3(h)Temporary Photo Library

Notification Panel / Alerts View

Displays real-time notifications about issue updates including case approvals, status changes, resolution updates, and new messages. This panel helps users stay informed about their reported issues and system activities with timely alerts and progress information.

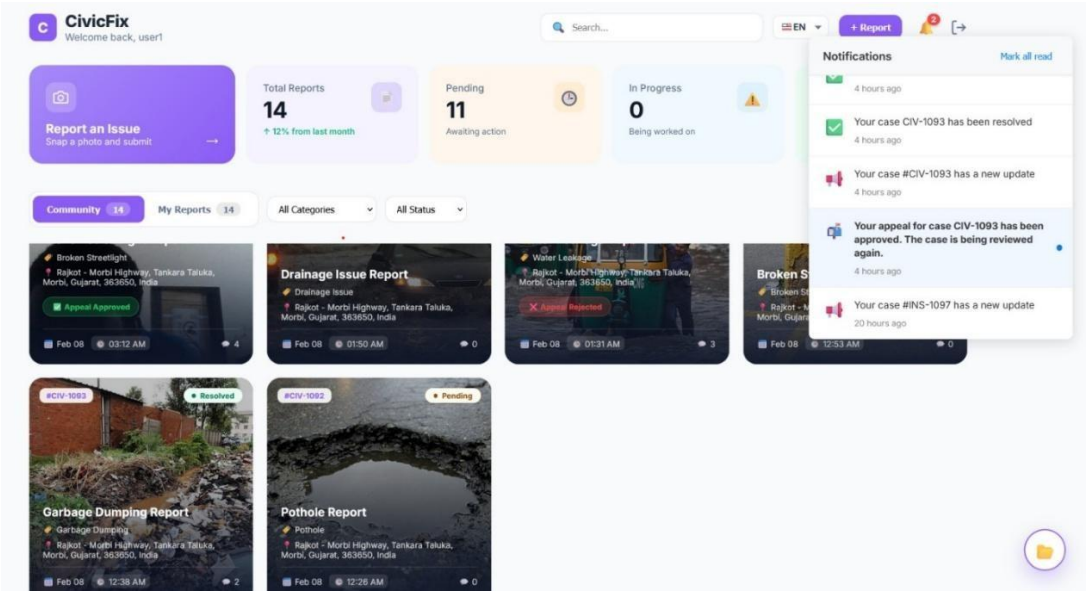

Fig 3(i)Notification Panel

4. Conclusion

4.1 Summary of the Results

The Crowdsourced Civic Issue Reporting and Resolution System successfully meets the objectives defined at the beginning of the project[3],[4]. The results obtained after implementation and testing clearly show that the system provides an effective digital platform for reporting, managing, and resolving civic issues in a structured and transparent manner.

Through the developed system, citizens were able to register, log in, and report civic issues easily using a web-based interface. The complaint submission process, which includes issue description, image upload, and location details, worked smoothly and ensured that problems were recorded accurately. The ability to track complaint status in real time helped users stay informed and reduced the need for repeated follow-ups.

From the administrative side, the system proved useful in managing civic complaints efficiently. Administrators were able to verify issues, assign them to appropriate staff members, and monitor their progress through a centralized dashboard. Staff members could view assigned tasks, update issue status, and provide resolution details, which improved coordination and accountability[6].

The system also demonstrated effective data storage and reporting capabilities. All complaints and their resolution histories were stored securely in the database, allowing authorities to review past records and analyze recurring civic issues. Overall, the results indicate that the proposed system improves efficiency, transparency, and communication in civic issue management when compared to traditional manual methods[6].

4.2 Advantages of the Proposed Work / Results / Methodologies

The proposed Crowdsourced Civic Issue Reporting and Resolution System offers several advantages over traditional manual methods of handling civic complaints. One of the major benefits of the system is the ease with which citizens can report issues. By providing a simple web-based platform, the system removes the need for physical visits or repeated phone calls, making the complaint process more convenient and accessible for everyone[1].

Another important advantage is improved transparency in the issue resolution process. The ability for users to track the status of their complaints in real time helps build trust between citizens and civic authorities. Citizens are kept informed at every stage, which reduces uncertainty and frustration commonly associated with manual complaint systems[4].

The system also improves efficiency for administrators and staff members. With a centralized dashboard, administrators can easily review, verify, and assign complaints to appropriate staff. This structured workflow helps reduce delays and ensures that issues are handled in an organized manner[1],[5]. Staff members benefit from clear task assignments and detailed complaint information, allowing them to take faster and more effective action.

From a data management perspective, the system provides a significant advantage by storing all civic issue data in a centralized and structured database. This organized data can be used to generate reports, identify recurring problems, and support better decision-making for future civic planning. Proper data storage also reduces the chances of data loss and duplication.

The use of role-based access control enhances system security by ensuring that users, administrators, and staff members can access only the features relevant to their roles. This

protects sensitive information and maintains the integrity of the system. Overall, the proposed methodology delivers a reliable, scalable, and user-friendly solution that improves civic engagement and service delivery[5].

4.3 Scope of Future Work

Although the Crowdsourced Civic Issue Reporting and Resolution System successfully meets its current objectives, there is significant scope for future enhancements that can further improve its effectiveness and usability[4]. One possible extension of the system is the development of a dedicated mobile application for Android and iOS platforms. A mobile app would allow citizens to report issues more conveniently using smartphones and enable features such as real-time location detection through GPS.

In the future, advanced features such as automatic issue categorization and priority detection can be integrated using machine learning techniques. By analyzing complaint descriptions and images, the system could automatically identify urgent issues and assign higher priority to critical problems, helping authorities respond more quickly to serious concerns[1].

The system can also be enhanced by integrating Geographic Information System (GIS) mapping. This would allow civic authorities to visualize reported issues on a map, making it easier to identify problem-prone areas and plan targeted improvements. Map-based visualization would also help in better resource allocation and long-term civic planning[2,],[6].

Another potential improvement is the integration of notification services such as SMS or email alerts[6]. This would ensure that users receive updates even when they are not actively using the application. Feedback and rating features can also be added, allowing citizens to provide feedback on issue resolution quality, which can help improve service performance.

Overall, the scope of future work highlights the flexibility and scalability of the proposed system. With additional features and technological enhancements, the system can evolve into a more intelligent, responsive, and comprehensive civic engagement platform[5].

4.4 Unique Features of the Proposed System

The Crowdsourced Civic Issue Reporting and Resolution System includes several unique features that distinguish it from traditional civic complaint handling methods[4]. One of the most notable features is the direct involvement of citizens in reporting and monitoring civic issues. By allowing users to submit complaints online and track their progress in real time, the system encourages active public participation and strengthens civic engagement.

Another unique feature of the system is its transparent workflow. Each reported issue follows a clearly defined process, starting from submission and verification to assignment and resolution. Users can view status updates at every stage, which helps build trust and reduces uncertainty regarding complaint handling[2].

The system also stands out due to its role-based access structure. Separate dashboards for citizens, administrators, and staff members ensure that each user interacts with the system according to their responsibilities. This organized approach improves efficiency, accountability, and security within the platform[1].

The inclusion of image upload functionality is another important feature that adds value to the system. Visual evidence helps authorities understand issues more accurately and reduces false or unclear reporting. This leads to quicker verification and better decision-making.

Additionally, the centralized data management feature allows all civic issue information to be stored in a structured database. This not only supports efficient complaint handling but also enables future analysis of recurring issues and affected areas. Overall, these unique features make the proposed system more effective, user-friendly, and reliable compared to conventional methods[1],[3].

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Appendix

Appendix A

Summary of Key Information

• Project Name:

Crowdsourced Civic Issue Reporting and Resolution System

• Purpose:

To design and develop a centralized, web-based platform that allows citizens to report civic issues digitally and enables municipal authorities to manage, track, and resolve these issues efficiently. The system aims to reduce manual complaint handling, improve transparency, ensure accountability, and enhance communication between citizens and civic authorities.

• Key Technologies Used:

1. Web-based Frontend (HTML, CSS, JavaScript)
2. Backend Services (PHP)
3. Database Management System (MySQL)
4. Web Server Environment (Apache using XAMPP/WAMP)
5. Role-Based Authentication and Session Management
6. Image Upload and Evidence Handling Module

• Target Users:

1. **Citizens:**
   To register, report civic issues, upload images, and track complaint status.
2. **Administrators:**
   To verify complaints, assign issues to staff members, monitor progress, and manage system activities.
3. **Staff Members:**
   To view assigned complaints, take necessary action, and update issue status.

• Key Features and Innovations:

1. Online civic issue reporting with image and location support.
2. Real-time complaint status tracking for citizens.
3. Role-based dashboards for users, administrators, and staff.
4. Centralized database for structured complaint management.
5. Transparent workflow from issue submission to resolution.
6. Improved accountability through clear task assignment.

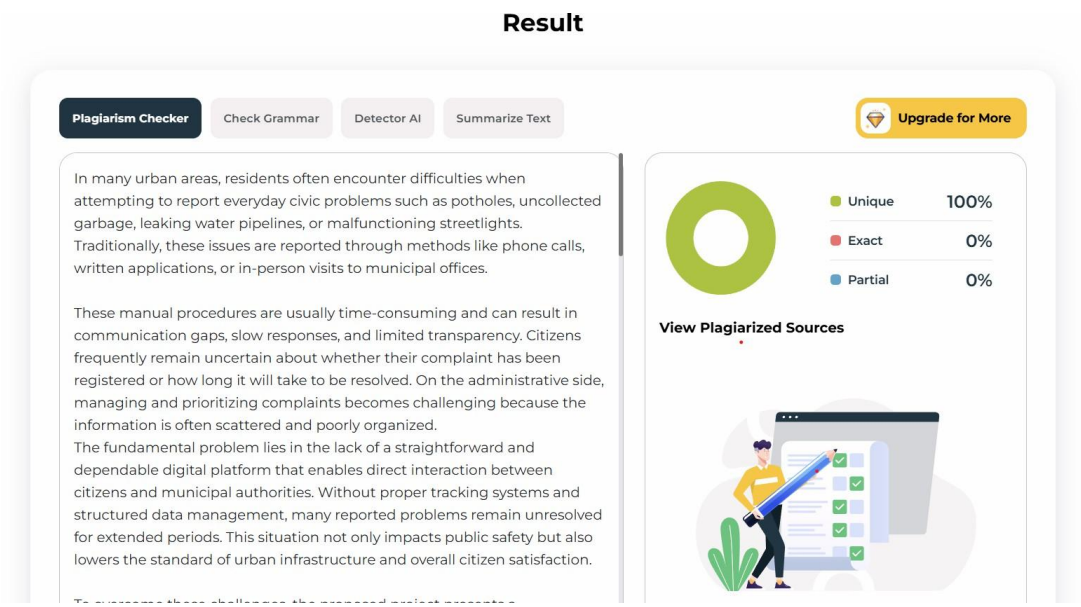
• Current Status:

A fully functional prototype of the Crowdsourced Civic Issue Reporting and Resolution System has been successfully developed and tested. Core functionalities such as user registration, issue reporting, admin verification, staff assignment, status tracking, and database storage have been implemented and validated through testing.

• Future Enhancements:

1. Development of a mobile application for easier citizen access.
2. Integration of GIS-based map visualization for issue locations.
3. Automated issue prioritization based on severity and category.
4. SMS and email notification support for status updates.
5. Advanced analytics for identifying recurring civic issues and affected areas.

• Supporting Materials:

1. Citizen, admin, and staff dashboards
2. Civic issue reporting and tracking interfaces
3. Complaint status reports and history records
4. Database tables and stored complaint data
5. System architecture and workflow diagrams
6. Project documentation and screenshots

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